

One-Pass Deep Solvers for Stationary Hamilton–Jacobi–Bellman Equations via Convexity Constraints and Homotopy Continuation

Working paper — experimental report

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Abstract

Physics-informed training of neural networks on the residual of a stationary Hamilton–Jacobi–Bellman (HJB) equation reliably converges to *spurious solutions*: functions that satisfy the partial differential equation to high accuracy but are not the value function of the underlying optimal control problem. Prior work resolved this with a two-pass scheme, first fitting the network to data generated from a State-Dependent Riccati Equation (SDRE) approximation and then refining on the PDE residual. This paper investigates whether the data pass can be removed entirely. We show experimentally that (i) hard architectural convexity (input-convex neural networks, ICNNs) defines the correct solution class but is not trainable through the residual loss on our benchmarks; (ii) a soft hinge penalty on the smallest Hessian eigenvalue of the *output*, combined with a strong-convexity margin and a data-free equilibrium anchor, recovers the exact value function of a 2D linear-quadratic regulator (LQR) in a single pass (evaluation MSE 2×10^{-8}); and (iii) on a nonlinear benchmark the same stack stalls in an optimization plateau, which is eliminated by *homotopy continuation* in the nonlinearity parameter, recovering the true value function to MSE 6×10^{-6} — five orders of magnitude better than any non-continuation variant and three orders better than the SDRE warm start used as supervised data in the two-pass approach. The resulting recipe — resampled residual + convexity margin penalty + equilibrium anchor + homotopy continuation — uses no precomputed solution data anywhere, and scales beyond closed-form eigenvalues: with a directional-curvature penalty it solves a ten-dimensional nonlinear problem with provably convex value function to $\approx 1\%$ relative error, strictly convex at every audited point. We also delimit the method’s scope: a finite-difference study shows the Cucker–Smale flocking value function is *not* convex in the domain interior, so the convexity prior is misspecified there. Finally we show the soft penalty is not the only route: two architectures are convex *by construction* over unconstrained parameters and need no penalty — a repaired ICNN (input skip connections plus smooth positivity), where a controlled ablation shows the skip connection alone moves LQR accuracy from 1.8×10^{-2} to 2.2×10^{-7} , and a log-sum-exp-of-quadratics network, exact on LQR (3.6×10^{-9}). The ICNN’s original failure traces to weight-space constraint geometry, not to convexity itself. We document the full experimental trail, including negative results, diagnostics, and the failure modes of each ablated ingredient.

1 Introduction

The value function V of an infinite-horizon optimal control problem satisfies a stationary HJB equation. Solving this PDE with a neural network trained on the squared residual — the physics-informed neural network (PINN) paradigm [1] — is attractive because it avoids grid-based discretization. It suffers, however, from a structural problem: on a bounded domain and without boundary conditions, the stationary HJB equation does not have a unique solution [8]. Gradient

descent on the residual loss happily converges to solution branches that have nothing to do with the optimal control problem.

In earlier work [2] we addressed this with a *two-pass* scheme: pass 1 fits the network to values and gradients of an approximate solution obtained from the State-Dependent Riccati Equation (SDRE) [5], placing the network in the basin of the true solution; pass 2 minimizes the PDE residual to refine beyond the accuracy of the SDRE data. The scheme works, but it requires generating problem-specific data, and the SDRE machinery does not scale gracefully.

This paper asks: **can the data pass be replaced by structural knowledge alone?** The structural facts available for a broad class of problems are:

1. the value function is *convex* (e.g. LQR; and verified numerically for our nonlinear benchmark, Section 5.1);
2. the origin is the equilibrium: $V(0) = 0$, $\nabla V(0) = 0$;
3. the problem family often interpolates continuously to a solvable base case (e.g. a nonlinearity parameter ε with $\varepsilon = 0$ giving LQR).

None of these requires precomputing a solution.

Contributions.

1. A systematic negative result: skip-free ICNNs — convex by construction — cannot be trained through the HJB residual on our benchmarks. We trace the failure to two distinct mechanisms (dying units under weight clipping; a universal degenerate attractor) and show it persists across activations, initializations and training budgets (Section 3).
2. A soft *output-convexity* formulation: a differentiable hinge penalty on $\lambda_{\min}(\nabla^2 V_\theta(x))$ with a strong-convexity margin, which acts as a solution-selection mechanism and solves the LQR benchmark exactly in one pass (Section 4).
3. The identification of a residual optimization plateau on the nonlinear benchmark — after eliminating, via targeted diagnostics, every competing explanation (feasibility, capacity, collocation overfitting, evaluation artefacts) — and its resolution by homotopy continuation in the nonlinearity (Section 5).
4. A scaling of the recipe to ten dimensions: a directional-curvature penalty replaces the closed-form 2D eigenvalue, solving a 10D nonlinear problem with provably convex value function to $\approx 1\%$ relative error, strictly convex at every audited point; and a scope result showing the Cucker–Smale flocking value function is non-convex in the domain interior, so the convexity prior does not apply there (Section 6).
5. Two certified, penalty-free alternatives, with a diagnosis of why the skip-free ICNN fails (weight-space constraint geometry, not the convexity property): (a) a *repaired ICNN* — input skip connections plus smooth positivity — where a controlled ablation shows the skip connection alone moves LQR from 1.8×10^{-2} to 2.2×10^{-7} ; and (b) a log-sum-exp-of-quadratics architecture, m -strongly convex by construction over unconstrained parameters and exact on LQR (3.6×10^{-9}). Both solve all benchmarks convex-by-construction with no penalty term (Section 7).
6. A complete ablation trail showing each ingredient of the final recipe is necessary, together with a candid pros-and-cons analysis of all approaches tried (Section 8).

2 Problem setting

We consider control-affine dynamics and quadratic control cost,

$$\dot{y} = f(y) + g(y)u, \quad J = \int_0^\infty (\ell(y(t)) + \beta \|u(t)\|^2) dt, \quad (1)$$

whose value function satisfies the stationary HJB equation

$$\mathcal{R}[V](x) := \langle \nabla V(x), f(x) \rangle - \frac{1}{4\beta} \nabla V(x)^\top g(x) g(x)^\top \nabla V(x) + \ell(x) = 0, \quad (2)$$

with optimal feedback $u^*(x) = -\frac{1}{2\beta} g(x)^\top \nabla V(x)$. Note that (2) involves only ∇V ; the value function is determined up to an additive constant, which we fix by evaluation-time shift correction or by the equilibrium anchor below.

Benchmarks. Two problems on the domain $[-1, 1]^2$:

- **LQR2D:** $f(x) = Ax$, $g = B$, $\ell(x) = \frac{1}{2}x^\top Qx$, with known analytic solution $V^*(x) = \frac{1}{2}x^\top Px$, $P = \frac{1}{5}(1 + \sqrt{6})I$ (Hessian floor ≈ 0.69).
- **NonLinear2D:** $\dot{y}_0 = y_1$, $\dot{y}_1 = \varepsilon y_0^3 + u$ with $\varepsilon = 1$, $\beta = \frac{1}{2}$, $\ell(x) = \frac{1}{2}\|x\|^2$. Ground truth on a 100×100 grid from the prior work’s converged solver; values in $[0, 5.7]$.

Spurious solutions. On a bounded domain without boundary conditions, (2) admits multiple solutions; for LQR both Riccati branches solve the PDE, and only the stabilizing (convex) branch is the value function. This is the central difficulty: *a small residual does not certify the right solution*. Empirically, an unconstrained network trained on the residual of LQR2D reaches residual 4.7×10^{-8} while its evaluation error against the true value function remains at 4.6×10^{-2} (Table 1); the same phenomenon appears on NonLinear2D, where the expected residual reaches 2.3×10^{-5} at evaluation error 1.2 (Table 2), and the learned surface is visibly a different object from the value function (Figure 3).

Network, loss, and protocol. Unless stated otherwise the network is a multilayer perceptron with three hidden layers of width 32. The base loss is the mean squared residual at $n = 500$ interior collocation points. Training uses Adam with a staged learning-rate schedule ($10^{-2} \rightarrow 10^{-5}$, 10k–28k epochs). Evaluation reports the MSE against the ground truth on a 100×100 grid (shift-corrected for residual-only runs), and convexity of the *learned network* is verified post hoc by the exact autograd Hessian on a 20×20 grid: we report the global minimum eigenvalue and the fraction of grid points violating positive semi-definiteness.

3 Route 1: hard architectural convexity (ICNN)

Input-convex neural networks [3] guarantee convexity by construction: all weights after the first layer are constrained non-negative and activations must be convex and non-decreasing. We tested the skip-free variant with the principled initialization of [4], several positivity mechanisms (lazy clipping, exponential reparameterization, and their no-positivity ablations), and a sweep of convexity-safe activations (ReLU, Softplus, ELU, CELU, LeakyReLU).

Failure mechanism 1: dying units under clipping. With ReLU, the combination of non-negative weights and the negative biases installed by the convex initializer kills hidden units: in the trained network 25/32 units of the second layer and 31/32 units of the third layer are dead (pre-activation negative on the whole domain), and the loss gradient is exactly zero for every parameter except the output layer. Training stalls within a few hundred steps.

Failure mechanism 2: a universal degenerate attractor. Several distinct configurations (LeakyReLU; Softplus with standard initialization; a wider Softplus network) collapse to *bit-identical* final losses (1.57×10^{-1}) — the same degenerate, nearly-affine function.

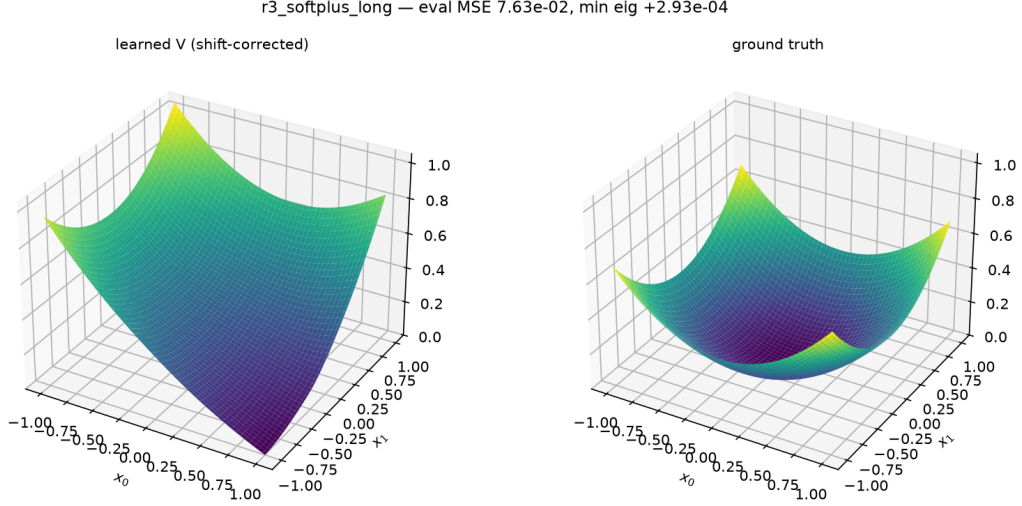


Figure 1: Best hard-ICNN configuration on LQR2D (Softplus, clipping, convex initialization, $3\times$ budget): strictly convex but underfitted (eval. MSE 7.6×10^{-2}) — the architectural route defines the right class but cannot navigate to the right element of it.

Best case still underfits. Smooth convex activations with the convex initializer do train and are *strictly* convex (positive minimum Hessian eigenvalue), but the residual plateaus near 6×10^{-3} — three orders above what unconstrained networks reach — and tripling the training budget changes nothing (Table 1, rows 5–6; Figure 1).

Remark 1. *Two caveats temper this negative result. First, a ReLU network is piecewise linear, so its Hessian is zero almost everywhere: the convexity check passes vacuously and the architecture contributes no useful curvature signal. Second, our ICNN lacks the input passthrough (skip) connections of the original architecture, which are also what make ICNNs universal over convex functions; the failure reported here is for the skip-free variant.*

4 Route 2: soft output convexity

Rather than constraining weights, we penalize non-convexity of the *output*. In 2D the smallest Hessian eigenvalue has a closed form: with $H = \nabla^2 V_\theta(x)$, $a = H_{11}$, $b = H_{12}$, $c = H_{22}$,

$$\lambda_{\min}(x) = \frac{1}{2}(a + c) - \sqrt{\frac{1}{4}(a - c)^2 + b^2}, \quad (3)$$

which is differentiable and assembled from one extra back-propagation through ∇V_θ (already computed for the residual). The training loss becomes

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{n} \sum_i \mathcal{R}[V_\theta](x_i)^2}_{\text{residual}} + w \underbrace{\frac{1}{n} \sum_i \max(0, m - \lambda_{\min}(x_i))}_{\text{convexity hinge, margin } m} + a \underbrace{(V_\theta(0)^2 + \|\nabla V_\theta(0)\|^2)}_{\text{equilibrium anchor}}. \quad (4)$$

The three ingredients encode, respectively: the PDE; convexity with a *strong-convexity margin* m (require $\lambda_{\min} \geq m$, chosen below the true solution’s curvature floor); and the equilibrium conditions $V(0) = 0$, $\nabla V(0) = 0$, which are problem knowledge, not data. Interior points are *resampled at every step*, so the loss targets the expected residual rather than a fixed collocation set. We also implemented an augmented-Lagrangian treatment of the constraint $\lambda_{\min}(x) \geq m$ (per-point multipliers on a fixed 20×20 grid) as an alternative to the fixed-weight hinge.

Table 1: LQR2D, one-pass residual-only training (seed 0 unless noted). “Viol.” is the fraction of test-grid points where the learned Hessian has a negative eigenvalue.

Configuration	Residual	Eval. MSE	$\min \lambda_{\min}$	Viol.
Unconstrained MLP (Sigmoid)	1.5×10^{-5}	4.4×10^{-2}	-0.46	100%
Unconstrained MLP (GELU)	4.7×10^{-8}	4.6×10^{-2}	-0.29	100%
ICNN, ReLU + clipping	1.6×10^{-1}	7.7×10^{-2}	0.0	0% (vacuous)
ICNN, Softplus + exp. weights	5.7×10^2	9.7×10^{-1}	+1.05	0%
ICNN, CELU + clipping + convex init	6.6×10^{-3}	9.1×10^{-2}	$+1.9 \times 10^{-4}$	0%
ICNN, Softplus, $3 \times$ budget	6.0×10^{-3}	7.6×10^{-2}	$+2.9 \times 10^{-4}$	0%
Hinge penalty $w=1, m=0$	1.6×10^{-4}	4.9×10^{-3}	-1.67	2%
Hinge penalty $w=1, m=0.1$	1.9×10^{-7}	1.7×10^{-7}	+0.60	0%
same, seed 1 / seed 2	—	$2.7 \times 10^{-6} / 7.7 \times 10^{-8}$	> 0	0%
+ equilibrium anchor	2.8×10^{-7}	2.1×10^{-8}	+0.62	0%

4.1 Results on LQR2D

Table 1 summarizes the progression. The decisive ingredient is the margin: the plain hinge ($m = 0$) improves the evaluation error tenfold over the unconstrained baseline, but the margin $m = 0.1$ — comfortably below the true curvature floor 0.69, excluding the concave spurious branch entirely — recovers the analytic Riccati solution to machine-practical accuracy: evaluation MSE 1.7×10^{-7} , strictly convex on the whole domain. Robust across seeds (2.7×10^{-6} , 7.7×10^{-8}), and the equilibrium anchor improves it further to 2.1×10^{-8} . Figure 2 shows the learned surfaces.

5 The nonlinear benchmark: diagnostics and continuation

Transferring the stack to NonLinear2D initially failed: every configuration — unconstrained, hinge, augmented Lagrangian, with or without anchor — landed at evaluation error ≈ 1.2 . Before changing the method we eliminated the competing explanations one by one.

5.1 Is the convex ansatz even well-specified?

A finite-difference Hessian analysis of the ground-truth grid shows the true value function *is* convex — minimum eigenvalue ≥ 0.02 everywhere, median 0.73. (Our first margin guess $m = 0.05$ exceeded the floor 0.02 in places and was infeasible; all subsequent runs use $m = 0.01$.)

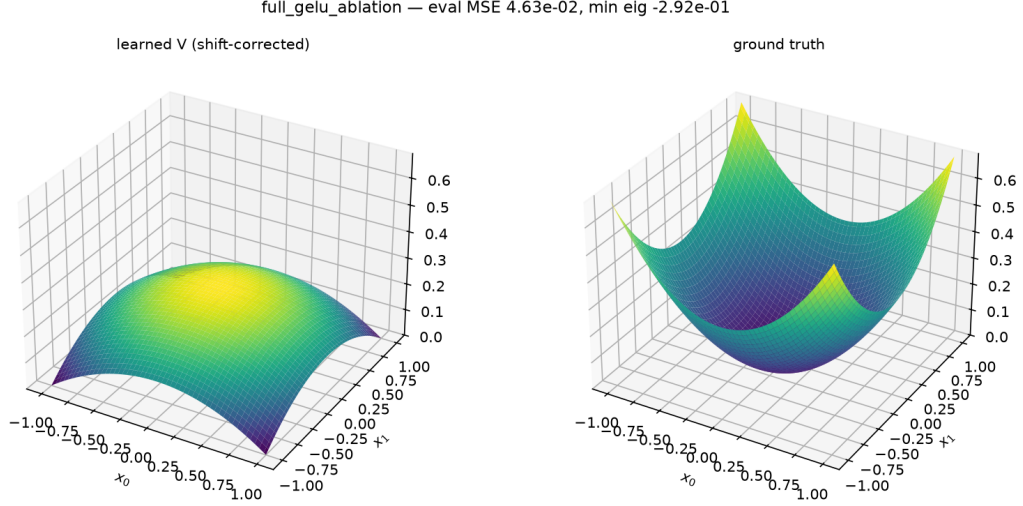
5.2 Is the evaluation sound?

A supervised control fit exposed a long-standing **evaluation bug**: the ground-truth grid was flattened without transposing, mismatching the orientation of the evaluation points; every historical evaluation error on this benchmark was computed against a transposed surface. We verified the correct orientation against the analytic SDRE formula (mismatch MSE 0.108 as stored vs 0.0076 transposed), fixed the loader, and re-scored all saved models. After the fix, the supervised fit scores 8.6×10^{-3} , essentially the SDRE’s own 7.6×10^{-3} gap to the truth (the small excess is network fit error), confirming the pipeline. The residual-only runs remain at ≈ 1.2 : genuinely spurious.

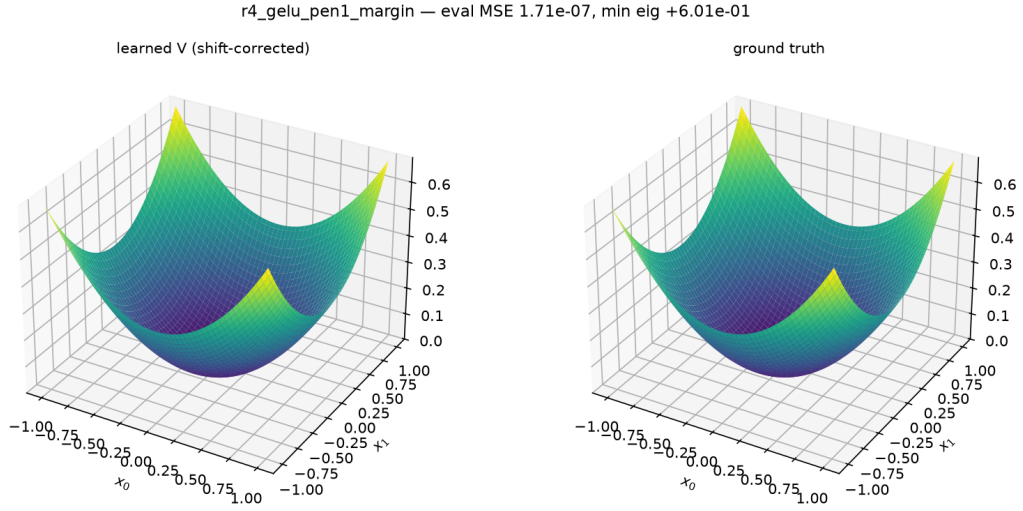
5.3 Is it feasible, representable, or overfitted?

Three facts pin the failure down to optimization:

- **Feasibility**: the ground truth satisfies the coded residual to MSE 3.4×10^{-7} (finite differences), so “convex + residual ≈ 0 + anchored” is achievable.



(a) Unconstrained GELU network: residual 4.7×10^{-8} , yet the learned surface (left) is a spurious solution, far from the truth (right).



(b) Same network with the convexity hinge and margin $m = 0.1$: the exact Riccati value function is recovered (eval. MSE 1.7×10^{-7}).

Figure 2: LQR2D learned value functions (left panels) vs ground truth (right panels).

- **Capacity:** the same architecture fits the ground truth *supervised* to MSE 3.7×10^{-6} ; the truth-fitted network’s own residual is 1.2×10^{-3} .
- **Collocation overfitting:** the unconstrained network’s residual degrades $100\times$ off its fixed 500 training points ($5 \times 10^{-6} \rightarrow 5 \times 10^{-4}$). Per-step resampling removes the overfit but does *not* remove the spurious branch: with resampling the expected residual still reaches 2.3×10^{-5} at evaluation error 1.20 — the spurious branch is a genuine alternative (near-)solution of the PDE, not an artefact.

Under the convexity constraint, every optimizer variant (fixed hinge, augmented Lagrangian, data-free warm shaping to a quadratic bowl, all budgets) converges to the *same* convex non-solution: residual stuck at $\approx 3 \times 10^{-2}$, minimum eigenvalue pinned at the margin (Figure 3, middle row). The truth (curvature floor 0.02, residual 3.4×10^{-7}) is feasible and strictly inside the constraint — the obstacle is purely the optimization landscape.

Table 2: NonLinear2D, corrected evaluation. All rows except the supervised control are one-pass residual-only (no solution data). “Stack” = resampled residual + hinge $w=10$, $m=0.01$ + anchor $a=10$.

Configuration	Residual	Eval. MSE	$\min \lambda_{\min}$	Viol.
Supervised control (fits SDRE data)	—	8.6×10^{-3}	-0.53	4%
Unconstrained, resampled	2.3×10^{-5}	1.20	-27.9	84%
Stack, no continuation	3.4×10^{-2}	1.25	$+9.9 \times 10^{-3}$	0%
Stack, augmented Lagrangian	3.3×10^{-2}	1.25	$+8.7 \times 10^{-3}$	0%
Stack + warm shaping	2.6×10^{-2}	1.19	$+7.9 \times 10^{-3}$	0%
Stack + ε-continuation	1.1×10^{-5}	5.9×10^{-6}	$+0.0202$	0%
AL variant + ε -continuation	5.6×10^{-6}	8.9×10^{-6}	$+0.0098$	0%

5.4 Homotopy continuation in the nonlinearity

The problem family interpolates to a solvable base case: at $\varepsilon = 0$ the dynamics are linear and the problem is an LQR, which the one-pass stack solves exactly. We therefore anneal ε from 0 to 1 over the first 12k of 28k training steps, tracking the continuous solution path $V(\varepsilon)$ — a numerical-continuation strategy [7] — instead of attacking the $\varepsilon = 1$ landscape cold.

The effect is decisive (Table 2): evaluation error drops from 1.2 to 5.9×10^{-6} (hinge variant) and 8.9×10^{-6} (augmented-Lagrangian variant) — five orders of magnitude — and the learned curvature floor ($+0.0202$) lands exactly on the measured floor of the true solution (0.02). Figure 3 (bottom) shows the learned surface is visually indistinguishable from the truth.

6 Beyond two dimensions: scope and scaling

6.1 When the convexity prior does not apply: Cucker–Smale

The Cucker–Smale flocking model (state = particle positions and velocities; interaction kernel $a(y_i, y_j) = (1 + (y_i - y_j)^2)^{-1}$; consensus cost) is the high-dimensional benchmark of our earlier work, and the natural next target. Before applying the method we asked whether its value function is convex at all. A 2024 study had found non-convex points of an SDRE-rollout approximation of V , but only near the domain boundary and with a fragile one-sided stencil — possibly artefacts.

We redid the analysis on the interior: 150 random points in $[-2, 2]^4$ (two particles, 4D state, domain $[-3, 3]^4$), central differences with $h = 0.05$ on the SDRE-rollout value. The result is unambiguous: **4.7% of interior points are non-convex, with minimum Hessian eigenvalue down to -0.25** against a median curvature of $+0.34$, and the violations lie well inside the domain ($|\text{positions}| \leq 1.3$, $|\text{velocities}| \leq 1.8$). The worst point pairs separated positions with opposing velocities — precisely where the non-convex interaction kernel is active. (Caveat: the SDRE rollout approximates V ; this is strong evidence, not proof.)

Two consequences. First, the convexity prior is *misspecified* for Cucker–Smale on this domain: a method that enforces convexity cannot converge to its value function, which retroactively explains why the 2024 ICNN experiments on this problem could not have succeeded regardless of trainability. Second, the method’s scope claim must be: problems whose value function is convex on the domain of interest — a property that, as here, can be checked numerically in advance. Extensions to Cucker–Smale would require a convexity-region restriction (e.g. near the consensus manifold) or a convex-plus-correction decomposition; both are open.

Table 3: 10D benchmark, one-pass residual-only (no solution data). Eval. MSE against the quadrature ground truth on 5000 random points; Hessian statistics from exact eigendecompositions at 300 random points. Typical value magnitude is ≈ 3 , so MSE 1.7×10^{-3} corresponds to $\approx 1\%$ relative error.

Configuration	Residual	Eval. MSE	min $\lambda_{\min}$	Viol.
Unconstrained (no penalty/anchor/continuation)	7.0×10^{-6}	1.53	-1.03	100%
Full stack (dir. curvature, $m=0.1$, anchor, continuation)	4.0×10^{-3}	1.7×10^{-3}	+0.61	0%

6.2 A ten-dimensional benchmark with provably convex value function

To test the recipe in higher dimension on a *well-specified* problem, we construct one whose value function is provably convex with an exact ground truth. Take coordinate-wise dynamics and cost

$$\dot{y}_i = \varepsilon y_i^3 + u_i, \quad \ell(y) = \frac{1}{2} \|y\|^2, \quad \beta = \frac{1}{2}, \quad y \in [-1, 1]^{10}. \quad (5)$$

The HJB equation decouples into scalar problems with closed-form stabilizing solution

$$v'(s) = 2\beta\varepsilon s^3 + s\sqrt{4\beta^2\varepsilon^2 s^4 + 2\beta}, \quad v''(s) = 6\beta\varepsilon s^2 + \sqrt{\cdot} + \frac{8\beta^2\varepsilon^2 s^4}{\sqrt{\cdot}} > 0, \quad (6)$$

where $\sqrt{\cdot}$ abbreviates the root $\sqrt{4\beta^2\varepsilon^2 s^4 + 2\beta}$ appearing in v' . Thus $V^*(x) = \sum_i v(x_i)$ is strictly convex with curvature floor $\sqrt{2\beta} = 1$ on all of $\mathbb{R}^{10}$ (margin $m = 0.1$ provably feasible), and the ground truth is available by one-dimensional quadrature. The network is a generic 10D MLP (64×3 , GELU); nothing in the method exploits the decoupling.

Two ingredients must scale. The closed-form 2×2 eigenvalue is replaced by a *directional-curvature hinge*: at each step, for each collocation point, draw $k = 4$ random unit directions v and penalize $\max(0, m - v^\top \nabla^2 V_\theta(x) v)$, where $v^\top H v$ costs one extra backpropagation through ∇V_θ . Since $v^\top H v \geq \lambda_{\min}$, with resampled points and directions this enforces the spectral constraint in expectation. The post-hoc audit computes full autograd Hessians and exact eigenvalues at 300 random points. Continuation anneals ε from 0 (a decoupled 10D LQR) to 1 over the first 12k of 28k steps, as in Section 5.

The spurious-solution phenomenon persists in ten dimensions: the unconstrained control drives the residual to 7×10^{-6} yet lands at evaluation error 1.53, non-convex at *every* audited point (Table 3, first row). The full stack solves the problem (Table 3, Figure 4): evaluation MSE 1.7×10^{-3} ($\approx 1\%$ relative), and the exact-eigenvalue audit finds the learned function *strictly* convex at every one of the 300 audited points, with minimum eigenvalue +0.61 — comfortably above the margin and below the true floor of 1, consistent with a slightly smoothed approximation of the true curvature. Accuracy is lower than in 2D (10^{-3} vs 10^{-6}), as expected from the fixed network width and the Monte Carlo nature of both the collocation and the curvature constraint; we did not tune the 10D run.

7 Certified architectures without a penalty

The soft penalty solves both 2D problems and the 10D problem, but it carries no hard guarantee: convexity is verified at sampled points after training, the margin is a tunable loss weight, and the convexity term competes with the residual during optimization. The ICNN had the opposite profile — a hard guarantee, but untrainable through the residual (Section 3). We now show these are not the only options: *two* architectures are *both* convex by construction *and* trainable with no penalty term at all. The first repairs the ICNN directly; the second changes the function class.

Table 4: Repaired ICNN (FICNN), trained on residual + anchor (+ continuation for the nonlinear cases) with *no* convexity penalty. The skip ablation (top block) holds everything else fixed.

Problem	Variant	Eval. MSE	min $\lambda_{\min}$	Viol.
LQR2D	FICNN, skip	2.2×10^{-7}	+0.66	0%
	FICNN, skip + $m=0.05$	1.8×10^{-7}	+0.66	0%
	FICNN, <i>no</i> skip (control)	1.8×10^{-2}	0.0	stalled
NonLinear2D	FICNN, skip (+ continuation)	4.1×10^{-3}	+0.38	0%
10D cubic	FICNN, skip (+ continuation)	1.8×10^{-1}	+0.10	0%

Diagnosis: why the ICNN constraint, not convexity, was the problem. The ICNN failure is often misattributed to convexity being “only local” — it is not: an ICNN is globally convex on all of $\mathbb{R}^d$, and our trained ICNNs were strictly convex on the entire test grid. The failure is the *geometry of the parameterization*. Enforcing $W \geq 0$ by clipping makes the feasible set a polyhedral cone whose faces trap weights and, under ReLU, kill units irreversibly (Section 3); non-negative weights bias signal propagation toward a single degenerate attractor; and the skip-free variant is not universal over convex functions. The constraint is a *sufficient* condition on weights (non-negativity) imposed on a parameter space hostile to gradient descent, rather than the *actual* property (a positive semi-definite Hessian) imposed on the output.

7.1 Fixing the ICNN: skip connections and smooth positivity

Each item in the diagnosis has an architecture-level remedy, applied without touching the loss. We take the fully-input-convex network (FICNN) of [3] and make three changes:

1. **Input skip connections** at every layer ($z_{l+1} = g(W_l^z z_l + W_l^x x + b_l)$, with $W_l^z \geq 0$ but W_l^x *unconstrained*). The passthrough of x is exactly what makes ICNNs universal over convex functions; the 2024 variant had dropped it.
2. **Smooth positivity by reparameterization**: the non-negative weight is softplus(ρ) for an unconstrained ρ . There is no clipping, hence no trapping faces, and gradients flow through every unit at every step.
3. **Smooth convex activation** ($g = \text{softplus}$), convex and non-decreasing with nowhere-zero derivative, so units cannot die as ReLU units do.

An optional $\frac{m}{2}\|x\|^2$ term makes the result m -strongly convex. Convexity holds for *any* values of the unconstrained parameters, so the parameter space is flat and Adam runs unimpeded.

The skip connection is decisive, and a controlled ablation isolates it (Table 4): on LQR2D, with everything else identical (smooth positivity, softplus activation, anchor), the skip-free FICNN stalls at evaluation error 1.8×10^{-2} while the skip version reaches 2.2×10^{-7} — five orders of magnitude, confirming that the 2024 skip-free failure (Section 3) was the missing passthrough, not the convexity constraint. With skips the repaired ICNN solves all three problems convex-by-construction and penalty-free (Figure 5): NonLinear2D to 4.1×10^{-3} (with continuation) and the 10D problem to 1.8×10^{-1} , both with zero convexity violations.

A capacity sweep, however, reveals the FICNN’s ceiling: on NonLinear2D, widening the network from 64 to 128 to 256 (depth 3) makes accuracy *worse*, not better ($4.1 \times 10^{-3} \rightarrow 1.5 \times 10^{-2} \rightarrow 2.3 \times 10^{-2}$; Figure 6, right), and adding a layer does not help either. The gap to the penalty method is therefore *not* a capacity limitation but a trainability one: the constrained-weight signal propagation conditions worse as the network grows — the well-documented difficulty of scaling ICNNs that motivated dedicated initialization schemes [4]. This contrasts sharply with the next architecture, whose accuracy improves with size.

7.2 Changing the class: log-sum-exp of quadratics

A second route abandons weight-sign constraints entirely and builds convexity into the function form.

Construction. Place the constraint in the output directly:

$$V_\theta(x) = \frac{m}{2}\|x\|^2 + \frac{1}{\tau} \log \sum_{i=1}^K \exp\left(\tau \left(\frac{1}{2}x^\top L_i L_i^\top x + b_i^\top x + c_i\right)\right), \quad (7)$$

with parameters $\{L_i \in \mathbb{R}^{d \times d}, b_i \in \mathbb{R}^d, c_i \in \mathbb{R}\}$ *unconstrained*. Each component quadratic is convex because $L_i L_i^\top \succeq 0$ for any L_i (no constraint needed); log-sum-exp is convex and non-decreasing, so the composition is convex; and the quadratic prefactor gives, with $p(x) = \text{softmax}(\tau q(x))$,

$$\nabla^2 V_\theta(x) = mI + \sum_i p_i(x) L_i L_i^\top + \tau \text{Cov}_p(L_i L_i^\top x + b_i) \succeq mI. \quad (8)$$

Thus V_θ is m -strongly convex on all of $\mathbb{R}^d$ *by construction*: the margin is now an architectural constant, not a loss term, and no convexity penalty is used at all. This is the smooth (log-sum-exp) form of a max-of-quadratics expansion — the function class of max-plus methods for HJB equations [6] — and the induced feedback law $u^* \propto g^\top \nabla V_\theta = g^\top (mx + \sum_i p_i(x)(L_i L_i^\top x + b_i))$ is a softmax-gated mixture of linear controllers, recovering an SDRE-like structure without being told to. At $K = 1$ it is exactly a quadratic, so LQR is representable and ε -continuation starts inside the class.

Results. We train (7) with the residual and equilibrium anchor only — no convexity penalty — using the same continuation schedule as before. Table 5 reports the outcome. On LQR2D the architecture is *exact*, reaching evaluation MSE 3.6×10^{-9} — the best result in this paper, better than the penalty method — with the learned Hessian matching the true Riccati value $P \approx 0.69I$. On the nonlinear and 10D problems it solves the problem with a hard certificate (zero violations by construction, confirmed by exact eigendecomposition).

Quadratic components, non-quadratic value functions. A single quadratic ($K = 1$) can only represent an LQR-type value function, so it is natural to ask whether (7) is limited to quadratic problems. It is not: the log-sum-exp of K quadratics is a smooth approximation of their *maximum*, and max-of-quadratics (indeed max-of-affine) is dense in the convex functions — the function class of max-plus methods [6]. The K -sweep on NonLinear2D, whose true value function is non-quadratic, makes this concrete (Figure 6, left): $K = 1$ stalls at evaluation error 3.3×10^{-2} (the pure-quadratic ansatz, as expected), and accuracy falls by more than two orders of magnitude as components are added — 3.0×10^{-3} ($K=4$), 1.5×10^{-4} ($K=16$), saturating near 2×10^{-4} ($K=64$). The nonlinear capability comes precisely from *combining* quadratics; the learned minimum curvature also descends toward the true floor (0.52 at $K=1$ toward 0.02), evidence the larger basis resolves genuine curvature rather than collapsing onto one piece. Figure 7 shows the surfaces.

Residual risks specific to (7): a softmax temperature τ that may need annealing to avoid gating collapse (another homotopy), and an $O(Kd^2)$ parameter count that motivates low-rank L_i in high dimension.

7.3 Which route?

Table 6 places the three convexity mechanisms side by side. All three give a convex value function; they trade accuracy against the strength of the guarantee. The soft penalty is the most

Table 5: LSE-of-quadratics architecture (7), trained on residual + anchor (+ continuation for the nonlinear cases) with *no* convexity penalty. Convexity holds by construction; “Viol.” is from exact eigendecomposition and is zero throughout. Penalty-method numbers from Tables 1–3 shown for comparison.

Problem	Architecture / method	Eval. MSE	min $\lambda_{\min}$	Viol.
LQR2D	LSE-quad ($K=8$)	3.6×10^{-9}	+0.69	0%
	(penalty + MLP, best)	2.1×10^{-8}	+0.62	0%
NonLinear2D	LSE-quad ($K=1$, quadratic)	3.3×10^{-2}	+0.52	0%
	LSE-quad ($K=16$, $\tau=2$)	1.5×10^{-4}	+0.28	0%
	(penalty + MLP, best)	5.9×10^{-6}	+0.02	0%
10D cubic	LSE-quad ($K=8$)	5.6×10^{-2}	+0.57	0%
	LSE-quad ($K=32$, $\tau=4$)	2.9×10^{-2}	+0.98	0%
	(penalty + MLP)	1.7×10^{-3}	+0.61	0%

Table 6: The three convexity mechanisms, eval. MSE against ground truth. Penalty: soft Hessian-eigenvalue hinge on an unconstrained MLP, convexity verified at samples. FICNN / LSE-quad: convex by construction (zero violations provable), trained with no penalty. All use the equilibrium anchor and, for the nonlinear cases, ε -continuation.

	Penalty + MLP (verified at samples)	FICNN (repaired ICNN) (certified)	LSE-quad (certified)
LQR2D	2.1×10^{-8}	2.2×10^{-7}	3.6×10^{-9}
NonLinear2D	5.9×10^{-6}	4.1×10^{-3}	1.5×10^{-4}
10D cubic	1.7×10^{-3}	1.8×10^{-1}	2.9×10^{-2}

accurate everywhere (it constrains an otherwise-unconstrained MLP, the most expressive model), but certifies convexity only at sampled points. The two architectures certify by construction — zero violations provable, no margin to tune — and the LSE-of-quadratics is the most accurate of them (exact on LQR), while the repaired ICNN is the most faithful to the standard convex-network literature. The architectures cost one to three orders of accuracy on value functions far from piecewise-quadratic, at the K and widths tested. The practical reading: use the penalty when raw accuracy dominates, an architecture when a hard certificate or interpretability does, or distil a penalty-trained solution into an architecture to obtain both.

8 Discussion: pros and cons of each route

8.1 Summary of approaches

Table 7 collects the trade-offs of every approach tried, from the published two-pass baseline through each ingredient of the one-pass recipe.

8.2 Why the pieces compose

The ablation trail supports a clean division of labor: *the convexity margin decides which solution; continuation provides the path to it; the anchor fixes the gauge; resampling makes the residual mean what it claims*. Removing any one ingredient reproduces a specific failure: no penalty $\rightarrow$ spurious branch at near-zero residual; no margin $\rightarrow$ partial convexity violations and a 4.9×10^{-3} floor; no continuation $\rightarrow$ the 3×10^{-2} plateau; no resampling $\rightarrow$ collocation overfitting.

8.3 Limitations

1. **Dimensionality.** The directional-curvature penalty scales the method to 10D (Section 6.2), but enforces the spectral constraint only in expectation; its sample complexity in still higher dimension, and the accuracy gap vs 2D (10^{-3} vs 10^{-6}), are unquantified.
2. **Margin selection.** The margin must lie below the true curvature floor. We used known floors (0.02 in 2D, 1 in 10D); without them one would anneal the margin or estimate the floor on the fly. A principled rule is open.
3. **Certificate vs accuracy.** The penalty method verifies convexity only at sampled points; the LSE-of-quadratics architecture (Section 7) gives a hard certificate but is less accurate on far-from-quadratic value functions at the K tested. Closing this gap — larger or low-rank K , τ annealing, or distilling a penalty-trained solution into the architecture for both accuracy and a certificate — remains open.
4. **Statistical strength.** Key comparisons rest on 1–3 seeds. The LQR result is confirmed on three seeds; the continuation and 10D results on one seed each (the former on two constraint mechanisms).
5. **Scope of the convexity prior.** The method targets problems whose value function is convex on the domain of interest — a property that can and should be checked in advance, as done for Cucker–Smale in Section 6.1, where it fails. For such problems a convexity-region restriction or a convex-plus-correction decomposition is needed; both are open.

9 Conclusion

The data pass of [2] can be removed. Its role — selecting the physically correct solution among the many functions that satisfy the stationary HJB residual — is taken over by structural priors that cost nothing to state: convexity with a strong-convexity margin, equilibrium anchoring, and homotopy from a solvable base problem. On both benchmarks the resulting one-pass solver is not merely competitive with the two-pass baseline; it is more accurate than the SDRE data that the two-pass scheme trains on, by three orders of magnitude on the nonlinear problem.

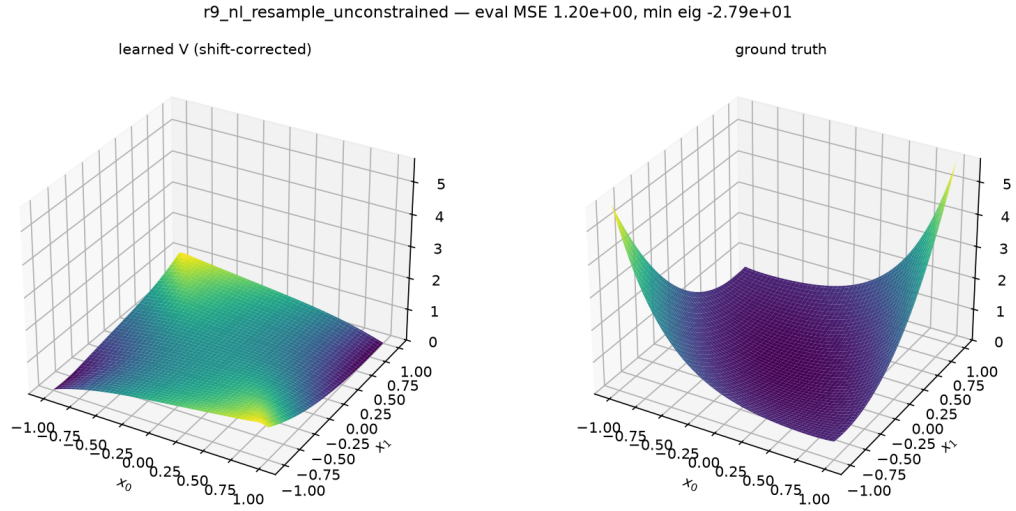
The recipe scales: with the directional-curvature penalty in place of the closed-form eigenvalue, a ten-dimensional nonlinear problem with provably convex value function is solved to $\approx 1\%$ relative error, strictly convex at every audited point. The scope is also now delimited: the Cucker–Smale value function is demonstrably non-convex in the domain interior, so flocking problems require an extension — a convexity-region restriction or a convex-plus-correction decomposition — rather than the method as-is.

Two routes deliver the convexity prior: a soft Hessian-eigenvalue penalty, most accurate but verified only at samples; and a log-sum-exp-of-quadratics architecture that is strongly convex by construction over unconstrained parameters, exact on LQR and certified everywhere, at some cost in accuracy. The architecture also clarifies the ICNN’s failure: convexity was never the obstacle (an ICNN is globally convex), the weight-space constraint geometry was.

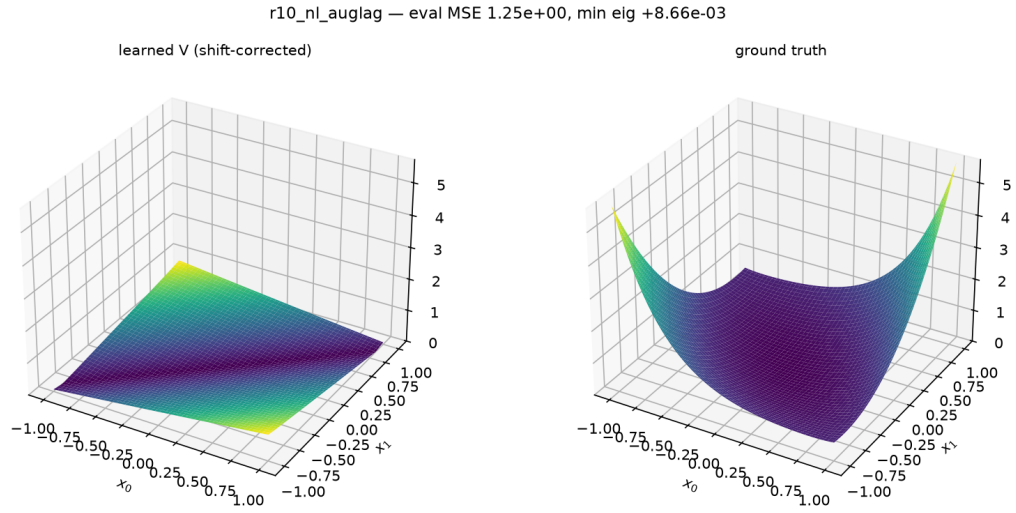
The immediate next steps are: a margin-selection rule; closing the architecture’s accuracy gap (low-rank or larger K , τ annealing, or distilling a penalty-trained solution into the architecture for accuracy and a certificate at once); the skip-connected ICNN control experiment; tuning and seed replication of the 10D result; and the convex-plus-correction extension toward Cucker–Smale. On the theory side, the empirical division of labor — convexity as solution selection, continuation as path-finding — invites a uniqueness result for strongly convex, anchored solutions of (2), and connects to continuation theorems studied in the companion work on Riemannian gradient smoothing (in preparation).

References

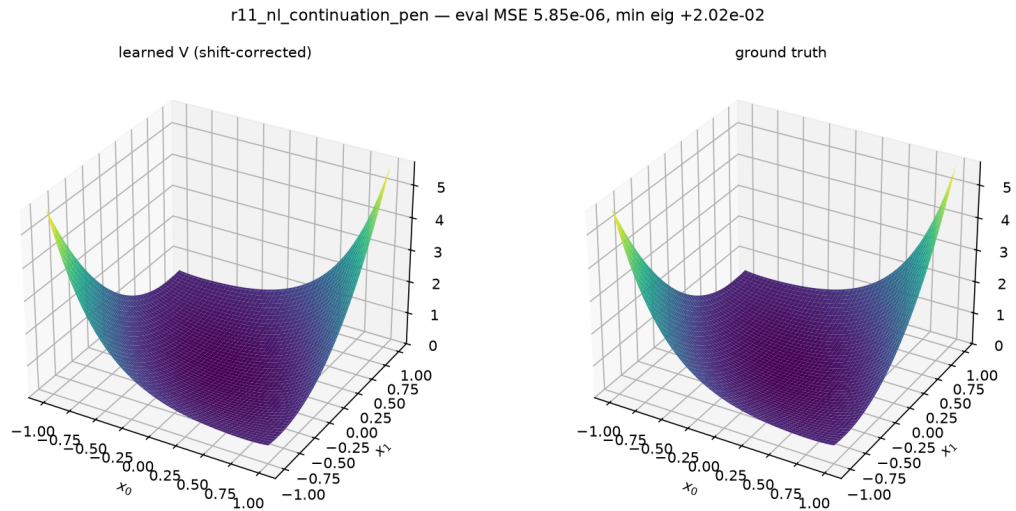
- [1] M. Raissi, P. Perdikaris, G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [2] A. Borovykh, D. Kalise, A. Laignelet, P. Parpas. Data-driven initialization of deep learning solvers for Hamilton–Jacobi–Bellman PDEs. *IFAC-PapersOnLine*, 55(30):168–173, 2022.
- [3] B. Amos, L. Xu, J.Z. Kolter. Input convex neural networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70:146–155, 2017.
- [4] P.-J. Hoedt, G. Klambauer. Principled weight initialisation for input-convex neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [5] H.T. Banks, B.M. Lewis, H.T. Tran. Nonlinear feedback controllers and compensators: a state-dependent Riccati equation approach. *Computational Optimization and Applications*, 37(2):177–218, 2007.
- [6] W.M. McEneaney. A curse-of-dimensionality-free numerical method for solution of certain HJB PDEs. *SIAM Journal on Control and Optimization*, 46(4):1239–1276, 2007.
- [7] E.L. Allgower, K. Georg. *Numerical Continuation Methods: An Introduction*. Springer, 1990.
- [8] M. Bardi, I. Capuzzo-Dolcetta. *Optimal Control and Viscosity Solutions of Hamilton–Jacobi–Bellman Equations*. Birkhäuser, 1997.



(a) Unconstrained, resampled: expected residual 2.3×10^{-5} , but the learned surface is a spurious branch (eval. MSE 1.20).



(b) Convexity-constrained without continuation: convex but stuck (residual 3×10^{-2} , eval. MSE 1.25).



(c) Full stack with ε -continuation: the true value function is recovered (eval. MSE 5.9×10^{-6}).

Figure 3: NonLinear2D learned value functions (left panels) vs ground truth (right panels).

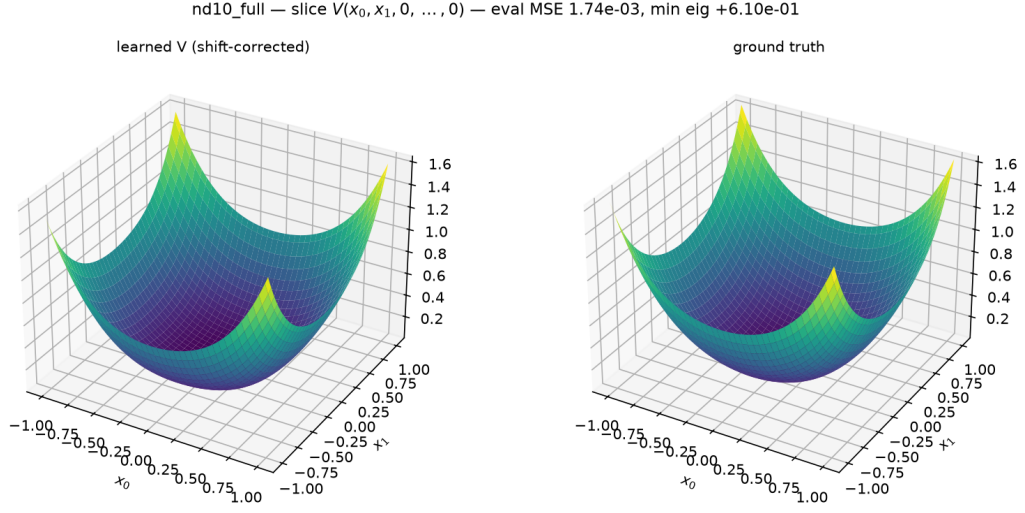


Figure 4: 10D benchmark: the slice $V(x_0, x_1, 0, \dots, 0)$ of the learned value function (left) against the ground truth (right).

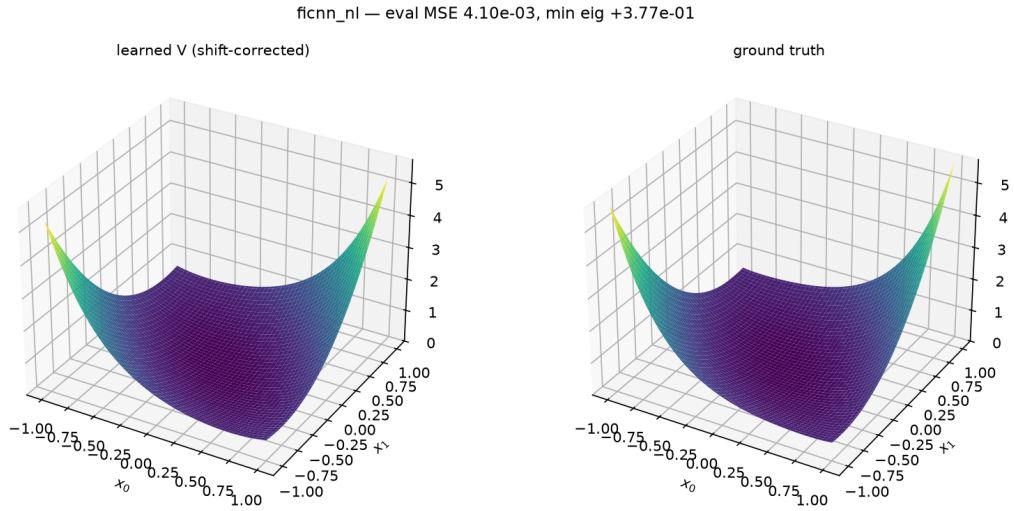


Figure 5: Repaired ICNN (FICNN with skip connections, smooth positivity) on NonLinear2D, trained with no convexity penalty: learned V (left) vs ground truth (right), eval. MSE 4.1×10^{-3} , convex by construction.

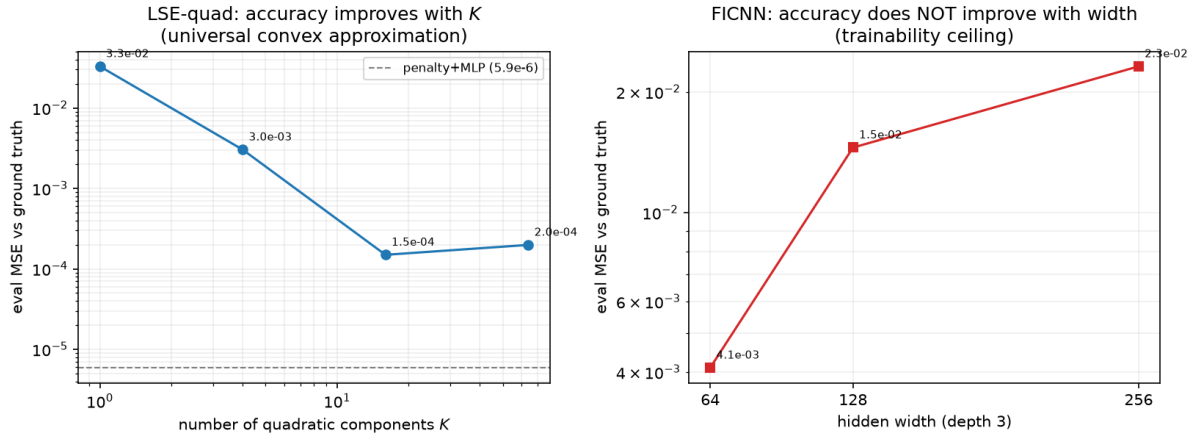
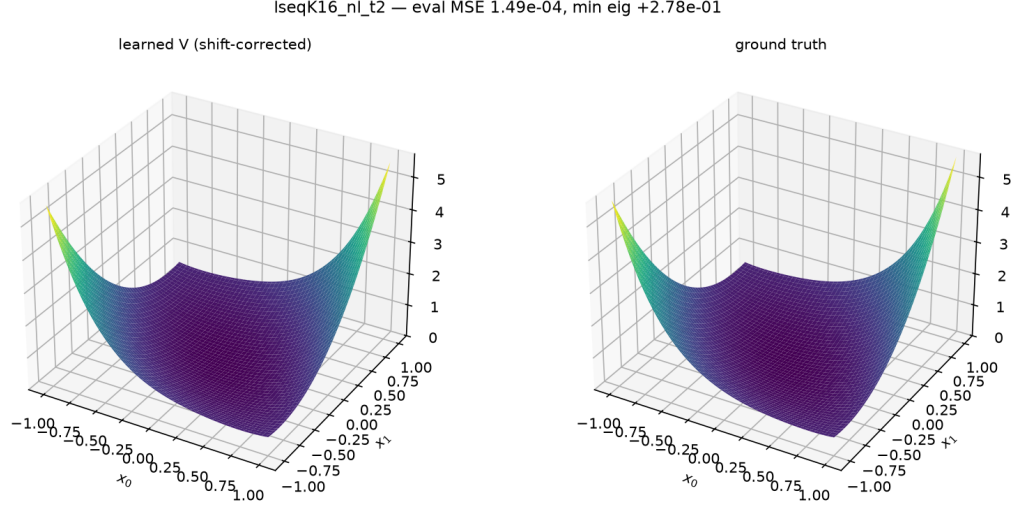
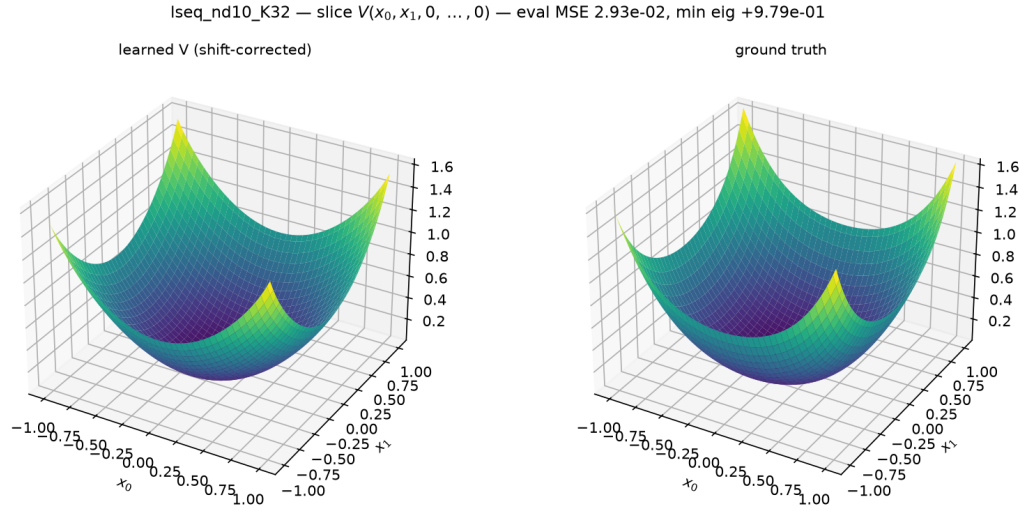


Figure 6: Capacity sweeps on NonLinear2D, both penalty-free and convex-by-construction. **Left:** LSE-quad accuracy improves by orders of magnitude as quadratic components K are added ($K=1$ is a single quadratic and fails), the signature of universal convex approximation. **Right:** the repaired ICNN does *not* improve with width — a trainability ceiling, not a capacity limit. Dashed line: penalty + MLP reference.



(a) NonLinear2D, $K = 16$: eval. MSE 1.5×10^{-4} .



(b) 10D, $K = 32$: slice $V(x_0, x_1, 0, \dots, 0)$, eval. MSE 2.9×10^{-2} .

Figure 7: LSE-of-quadratics architecture (left panels) vs ground truth (right panels), trained with no convexity penalty.

Table 7: Qualitative comparison of all approaches tried.

Approach	Pros	Cons
Two-pass (SDRE data $\rightarrow$ residual) [2]	Robust; data places the network in the right basin; published baseline.	Requires problem-specific data generation; SDRE itself is 7.6×10^{-3} from the truth here; two training stages.
Hard ICNN (skip-free)	Provable convexity certificate; zero violations by construction.	Not trainable through the residual on our benchmarks: dying units (ReLU), degenerate attractor, or a 6×10^{-3} residual plateau; ReLU variant passes the convexity check vacuously.
Repaired ICNN (FICNN: skip + smooth positivity)	Hard certificate; trainable (skip ablation: $1.8 \times 10^{-2} \rightarrow 2.2 \times 10^{-7}$ on LQR); stays within the standard convex-network family; unconstrained parameters.	Least accurate of the three convex methods on far-from-quadratic value functions (4×10^{-3} / 1.8×10^{-1} on NL2D / 10D).
LSE-of-quadratics architecture	Hard certificate <i>and</i> trainable (unconstrained parameters); exact on LQR (3.6×10^{-9}); margin is an architectural constant; interpretable mixture-of-feedbacks controller.	1–3 orders less accurate than penalty+MLP on far-from-quadratic value functions at the K tested; $O(Kd^2)$ parameters; τ may need annealing to avoid gating collapse.
Output hinge penalty + margin	Full expressivity of unconstrained nets; margin acts as solution selection; exact on LQR; simple (one weight, one margin).	No hard certificate (grid verification only); margin must lie below the true curvature floor, which may be unknown; $O(d^3)$ eigenvalue cost beyond closed-form 2D.
Augmented Lagrangian variant	Principled constraint handling; works comparably (8.9×10^{-6}).	More machinery (multiplier state, update schedule) with no measured gain over the fixed hinge.
Equilibrium anchor	Free (problem knowledge, not data); removes shift indeterminacy; improved LQR by an order of magnitude.	Insufficient alone — anchored unconstrained runs still pick spurious branches.
Warm shaping ($\frac{1}{2}\ x\ ^2$ pre-fit)	Data-free; cheap.	No measurable effect on the plateau.
Collocation resampling	Removes a genuine $100\times$ off-sample residual gap; standard	Does not by itself remove spurious branches.